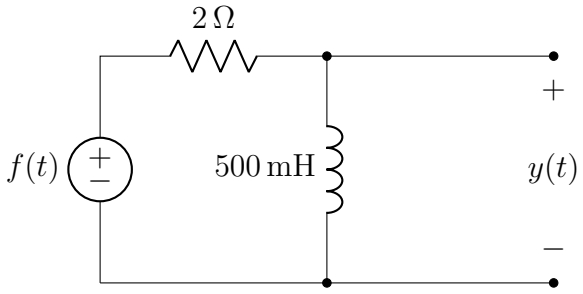
ECE 3210 hw09

1. Consider the circuit below.



(a) Derive the transfer function $H(s)$ of the circuit.

(b) For the input $f(t) = 3\sin(t - 60°)$, find the steady-state output $y(t)$.

(c) A periodic input has period $T = 2$ and is defined over one period by

$$f(t) = \begin{cases} 1, & 0 < t \leq 1, \\ 0, & 1 < t \leq 2. \end{cases}$$

Find the complex exponential Fourier series coefficients D_n, where

$$f(t) = \sum_{n=-\infty}^{\infty} D_n e^{j\pi nt}.$$

(d) Using the periodic input from the previous part, find the output Fourier series

$$y(t) = \sum_{n=-50}^{50} C_n e^{j\pi nt}.$$

Give the coefficients C_n.

Answer Key

1(a) $H(s) = \dfrac{s}{s+4}$

1(b) $y(t) = \dfrac{3}{\sqrt{17}} \sin\Big(t + \tan^{-1}(4) - \dfrac{\pi}{3}\Big)$

1(c) $D_0 = \dfrac{1}{2}, \quad D_n = \dfrac{1-(-1)^n}{j2\pi n}$ $(n \neq 0)$

1(d) $C_0 = 0, \quad C_n = 0$ for nonzero even $n, \quad C_n = \dfrac{4 - j\pi n}{16 + \pi^2 n^2}$ for odd n